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ENROLLMENT NUMBER : 23SS02IT181
SUBJECT NAME : Cryptography & Network Security
SUBJECT CODE : SSCS4020

Part A: Caesar Cipher

1. HELLO

Plaintext: HELLO

Ciphertext: KHOOR

Decryption: KHOOR → HELLO

2. WORLD

Plaintext: WORLD

Ciphertext: ZRUOG

Decryption: ZRUOG → WORLD

3. APPLE

Plaintext: APPLE

Ciphertext: DSSOH

Decryption: DSSOH → APPLE

4. INDIA

Plaintext: INDIA

Ciphertext: LQGLD

Decryption: LQGLD → INDIA

5. CYBER

Plaintext: CYBER

Ciphertext: FBEHU

Decryption: FBEHU → CYBER

6. SECURITY

Plaintext: SECURITY

Ciphertext: VHFXULWB

Decryption: VHFXULWB → SECURITY

7. NETWORK

Plaintext: NETWORK

Ciphertext: QHWZRUN

Decryption: QHWZRUN → NETWORK

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8. COMPUTER

Plaintext: COMPUTER

Ciphertext: FRPSXWHU

Decryption: FRPSXWHU → COMPUTER

9. ATTACK

Plaintext: ATTACK

Ciphertext: DWWDFN

Decryption: DWWDFN → ATTACK

10. DEFEND

Plaintext: DEFEND

Ciphertext: GHIHQQ

Decryption: GHIHQQ → DEFEND

Part B: Monoalphabetic Cipher

Plain	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	Q	W	E	R	T	Y	U	I	O	P	A	S	D	F	G	H	J	K	L	Z	X	C	V	B	N	M

1. HELLO

Plaintext: HELLO

Ciphertext: ITSSG

Decryption: ITSSG → HELLO

2. WORLD

Plaintext: WORLD

Ciphertext: VGKSR

Decryption: VGKSR → WORLD

3. APPLE

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Plaintext: APPLE

Ciphertext: QHHST

Decryption: QHHST → APPLE

4. INDIA

Plaintext: INDIA

Ciphertext: OFROQ

Decryption: OFROQ → INDIA

5. CYBER

Plaintext: CYBER

Ciphertext: ENWTK

Decryption: ENWTK → CYBER

6. ATTACK

Plaintext: ATTACK

Ciphertext: QZZQEA

Decryption: QZZQEA → ATTACK

7. DEFEND

Plaintext: DEFEND

Ciphertext: RTYTFR

Decryption: RTYTFR → DEFEND

8. COMPUTER

Plaintext: COMPUTER

Ciphertext: EGDHXZTK

Decryption: EGDHXZTK → COMPUTER

9. NETWORK

Plaintext: NETWORK

Ciphertext: FTZVGKA

Decryption: FTZVGKA → NETWORK

10. SECURITY

Plaintext: SECURITY

Ciphertext: LTEXKOZN

Decryption: LTEXKOZN → SECURITY

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Part C: Playfair Cipher (5 Marks)

Playfair Matrix (Keyword: MONARCHY)

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Q1. HELLO

Digraphs: HE | LX | LO

HE → CF
LX → SU
LO → PM

Ciphertext: CFSUPM

CF → HE
SU → LX
PM → LO

Decrypted Text: HELXLO

Q2. WORLD

Digraphs: WO | RL | DX

WO → VN
RL → MT
DX → BZ

Ciphertext: VNMTBZ

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VN → WO
MT → RL
BZ → DX

Decrypted Text: WORLDX

Q3. ATTACK

Digraphs: AT | TA | CK

AT → RS
TA → SR
CK → DE

Ciphertext: RSSRDE

RS → AT
SR → TA
DE → CK

Decrypted Text: ATTACK

Q4. DEFEND

Digraphs: DE | FE | ND

DE → CK
FE → GF
ND → RY

Ciphertext: CKGFRY

CK → DE
GF → FE
RY → ND

Decrypted Text: DEFEND

Q5. SECRET

Digraphs: SE | CR | ET

SE → LI
CR → DM
ET → KL

Ciphertext: LIDMKL

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LI → SE
DM → CR
KL → ET

Decrypted Text: SECRET

Q6. NETWORK

Digraphs: NE | TW | OR | KX

NE → MG
TW → QZ
OR → NM
KX → IZ

Ciphertext: MGQZNMIZ

MG → NE
QZ → TW
NM → OR
IZ → KX

Decrypted Text: NETWORKX

Q7. COMPUTER

Digraphs: CO | MP | UT | ER

CO → HM
MP → OL
UT → ZL
ER → KM

Ciphertext: HMOLZLKM

HM → CO
OL → MP
ZL → UT
KM → ER

Decrypted Text: COMPUTER

Q8. INDIA

Digraphs: IN | DI | AX

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IN → GA
DI → BK
AX → BA

Ciphertext: GABKBA

GA → IN
BK → DI
BA → AX

Decrypted Text: INDIA X

Q9. CYBER

Digraphs: CY | BE | RX

CY → HB
BE → CI
RX → AZ

Ciphertext: HBCIAZ

HB → CY
CI → BE
AZ → RX

Decrypted Text: CYBERX

Q10. INSTRUMENTS

Digraphs: IN | ST | RU | ME | NT | SX

IN → GA
ST → TL
RU → MZ
ME → CL
NT → RQ
SX → XA

Ciphertext: GATLMZCLRQXA

GA → IN
TL → ST
MZ → RU
CL → ME
RQ → NT
XA → SX

Decrypted Text: INSTRUMENTSX

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Part D: Vigenère Cipher (5 Marks)

Q1. HELLO

Answer:

Keyword: KEYKE

Ciphertext: RIJVS

Calculations:

- $H(7) + K(10) = R(17) \bmod 26$
- $E(4) + E(4) = I(8) \bmod 26$
- $L(11) + Y(24) = J(9) \bmod 26$
- $L(11) + K(10) = V(21) \bmod 26$
- $O(14) + E(4) = S(18) \bmod 26$

Decryption: RIJVS $\rightarrow$ HELLO

Q2. WORLD

Answer:

Keyword: KEYKE

Ciphertext: GSPVH

Calculations:

- $W(22) + K(10) = G(6) \bmod 26$
- $O(14) + E(4) = S(18) \bmod 26$
- $R(17) + Y(24) = P(15) \bmod 26$
- $L(11) + K(10) = V(21) \bmod 26$
- $D(3) + E(4) = H(7) \bmod 26$

Decryption: GSPVH $\rightarrow$ WORLD

Q3. APPLE

Answer:

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Keyword: KEYKE

Ciphertext: KTNVI

Calculations:

- $A(0) + K(10) = K(10) \bmod 26$
- $P(15) + E(4) = T(19) \bmod 26$
- $P(15) + Y(24) = N(13) \bmod 26$
- $L(11) + K(10) = V(21) \bmod 26$
- $E(4) + E(4) = I(8) \bmod 26$

Decryption: KTNVI $\rightarrow$ APPLE

Q4. INDIA

Answer:

Keyword: KEYKE

Ciphertext: SRBSE

Calculations:

- $I(8) + K(10) = S(18) \bmod 26$
- $N(13) + E(4) = R(17) \bmod 26$
- $D(3) + Y(24) = B(1) \bmod 26$
- $I(8) + K(10) = S(18) \bmod 26$
- $A(0) + E(4) = E(4) \bmod 26$

Decryption: SRBSE $\rightarrow$ INDIA

Q5. CYBER

Answer:

Keyword: KEYKE

Ciphertext: MCZOV

Calculations:

- $C(2) + K(10) = M(12) \bmod 26$
- $Y(24) + E(4) = C(2) \bmod 26$
- $B(1) + Y(24) = Z(25) \bmod 26$

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- $E(4) + K(10) = O(14) \bmod 26$
- $R(17) + E(4) = V(21) \bmod 26$

Decryption: MCZOV → CYBER

Q6. ATTACK

Answer:

Keyword: KEYKEY

Ciphertext: KXRKGI

Calculations:

- $A(0) + K(10) = K(10) \bmod 26$
- $T(19) + E(4) = X(23) \bmod 26$
- $T(19) + Y(24) = R(17) \bmod 26$
- $A(0) + K(10) = K(10) \bmod 26$
- $C(2) + E(4) = G(6) \bmod 26$
- $K(10) + Y(24) = I(8) \bmod 26$

Decryption: KXRKGI → ATTACK

Q7. DEFEND

Answer:

Keyword: KEYKEY

Ciphertext: NIDORH

Calculations:

- $D(3) + K(10) = N(13) \bmod 26$
- $E(4) + E(4) = I(8) \bmod 26$
- $F(5) + Y(24) = D(3) \bmod 26$
- $E(4) + K(10) = O(14) \bmod 26$
- $N(13) + E(4) = R(17) \bmod 26$
- $D(3) + Y(24) = B(1) \bmod 26$

Decryption: NIDORB → DEFEND

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Q8. NETWORK

Answer:

Keyword: KEYKEYKE

Ciphertext: XIRGSPBO

Calculations:

- $N(13) + K(10) = X(23) \bmod 26$
- $E(4) + E(4) = I(8) \bmod 26$
- $T(19) + Y(24) = R(17) \bmod 26$
- $W(22) + K(10) = G(6) \bmod 26$
- $O(14) + E(4) = S(18) \bmod 26$
- $R(17) + Y(24) = P(15) \bmod 26$
- $K(10) + K(10) = U(20) \bmod 26$
- (No 8th letter in keyword repetition? Actually keyword repeats as KEYKEYKE, so:)
- $K(10) + E(4) = O(14) \bmod 26$

Decryption: XIRGSPUO $\rightarrow$ NETWORK

Q9. COMPUTER

Answer:

Keyword: KEYKEYKE

Ciphertext: MSKNYROV

Calculations:

- $C(2) + K(10) = M(12) \bmod 26$
- $O(14) + E(4) = S(18) \bmod 26$
- $M(12) + Y(24) = K(10) \bmod 26$
- $P(15) + K(10) = Z(25) \bmod 26$
- $U(20) + E(4) = Y(24) \bmod 26$
- $T(19) + Y(24) = R(17) \bmod 26$
- $E(4) + K(10) = O(14) \bmod 26$
- $R(17) + E(4) = V(21) \bmod 26$

Decryption: MSKZYROV $\rightarrow$ COMPUTER

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Q10. SECURITY

Answer:

Keyword: KEYKEYKE

Ciphertext: CIASTGXW

Calculations:

- $S(18) + K(10) = C(2) \text{ mod } 26$
- $E(4) + E(4) = I(8) \text{ mod } 26$
- $C(2) + Y(24) = A(0) \text{ mod } 26$
- $U(20) + K(10) = E(4) \text{ mod } 26$
- $R(17) + E(4) = V(21) \text{ mod } 26$
- $I(8) + Y(24) = G(6) \text{ mod } 26$
- $T(19) + K(10) = D(3) \text{ mod } 26$
- $Y(24) + E(4) = C(2) \text{ mod } 26$

Decryption: CIAEVGDC → SECURITY